Mason Phounsavath

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EDUCATION

University of North Carolina - Chapel Hill *Bachelor's, Data Science, Computer Science Minor, Sigma Chi Fraternity* **Aug 2023 - May 2027**

Stanford University
Summer Semester Admit, Computer Science Concentration **Jun 2022 - Aug 2022**

PROFESSIONAL EXPERIENCE

Fidelity Investments | *Data Engineer Intern* **June 2026**

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PROJECTS

Bias Bars: Gender-Based Review Analysis | *Stanford University*

- Natural Language Processing (NLP) & Text Analytics - Extracted and analyzed word frequency distributions in professor reviews, applying tokenization and text normalization techniques.
- Data Processing & Statistical Computing - Implemented Python-based data pipelines to transform raw text into structured insights, leveraging dictionary-based aggregation and frequency normalization.
- Data Visualization & Interactive Dashboards - Developed an interactive Tkinter GUI for dynamic visualization of gender-based language patterns, using Plotlib-style rendering.
- Algorithm Optimization & Style Automation Engineering - Designed efficient data structures and vectorized operations to enhance computational efficiency in large-scale text processing.
- Software Development & Automation - Integrated file I/O operations, batch data parsing, and automated reporting, ensuring robust data ingestion and reproducibility.

2D Dungeon Crawler | UNC Chapel Hill

- Built a 2D dungeon crawler game using Java and JavaFX, implementing both front-end UI and back-end game logic.
- Designed and implemented interfaces and implementation classes to manage game state, including player movement, score tracking, and level progression.
- Integrated Object-Oriented Programming concepts such as encapsulation, interfaces, and polymorphism to structure maintainable code.

Financial Investment vs Performance in ACC Basketball | UNC Chapel Hill

- Built a predictive machine learning model using a decision tree regressor in Python (pandas, scikit-learn) to analyze the correlation between NCAA ACC men's basketball program funding and win percentage.
- Merged multi-source datasets containing team budgets, advanced metrics and performance stats for 15 ACC schools, performing extensive data cleaning, transformation, and visualization using matplotlib and seaborn.
- Achieved an R^2 score of 0.78, validating model accuracy in forecasting 2023 season outcomes based on 2022 financial and performance inputs.
- Provided actionable insights on resource allocation, ROI in athletics, and policy implications related to NIL-era college sports economics, using multiple data visualizations.

Skills

- **Programming Languages**: Python, JavaScript, HTML/CSS, SQL, MySQL, Java
- **Web Development**: React.js, Firebase Firestore, TailwindCSS, Component-Based Architecture
- **Data Science & Analytics**: Natural Language Processing (NLP), Statistical Computing, Data Visualization, Text Analytics, Data Analysis, Machine Learning, Statistical Insights,
- **Data Engineering**: Web Scraping (BeautifulSoup, Requests), Data Parsing, Frequency Normalization, Batch Data Processing, Tableau